Investigating the prevalence and determinants of HIV, HBV, HCV, and *Treponema pallidum* among blood donors in Douala, Cameroon

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1. Introduction

Blood safety is crucial for saving and preserving lives and improving the well-being of people in need around the world. Blood transfusion is a rapid and effective public health intervention for individuals with multi-factorial life-threatening anaemia [1]. However, blood transfusion remains a huge public health challenge in developing countries such as sub-Saharan Africa and Asian countries [2,3]. Although millions of human lives are saved every year globally, blood transfusion, if unsafe, can be a source of immunological adverse reactions and transfusion-transmissible infections (TTIs), which are mainly caused by viruses (e.g., hepatitis B and C viruses – HBV/HBC, Human immunodeficiency virus – HIV), bacteria (e.g., *Treponema pallidum*) and parasites (for example, *Plasmodium falciparum* and Babesia parasites) [4–6].

TTIs can exist asymptomatically in blood donors; thus, these donors should be screened for TTIs before blood transfusion [7]. African countries are facing rapid and high demand for blood product transfusion, and regions of the world where the TTIs burden is high and blood safety is particularly challenging [2,3]. TTIs are still a serious public health concern in Africa because they are responsible for enormous morbidity and mortality burdens alongside economic losses and are an important limiting factor to guaranteeing blood safety [4,5]. During the last two decades, many governments have made tremendous efforts to develop, implement, scale up, and improve their delivery systems and strategies for blood supply. This success was the result of several actions including recruitment of regular and volunteer non-remunerated blood donors, implementation of appropriate buildings, regular training and evaluation of health facility staff, systematic screening for major TTIs, effective quality control systems, and boosting funding [8].

According to the Cameroon National Blood Transfusion Program, 94,873 blood bags were collected in 2018, representing ~25% of the total need estimated at 400,000 blood bags (https://pnts-minsante.cm/accueil). TTIs are a major public health concern in the country. The government of Cameroon promulgated diverse laws in favour of better and safer blood transfusion systems and the management of blood transfusion-related complications [9]. Infectious diseases such as malaria, viral hepatitis, and HIV are highly prevalent in Cameroon [10–15], and their epidemiology is constantly changing, thereby justifying the need for continuous surveillance of TTI risk related to these diseases, to refine and/or optimize donor recruitment strategies. Studies documenting the epidemiology of TTIs in the country are lacking [16–23]. Therefore, in the present study, we determined the prevalence and determinants of TTIs in Douala, the most populated town in Cameroon.

2. Materials and Methods

2.1. Study design

A cross-sectional study was conducted from December 2021 to February 2022 at the blood transfusion centre (BTC) of the Douala Laquintinie Hospital (DLH) in the town of Douala, Littoral region, Cameroon. After obtaining administrative, research, and ethical authorizations, the study was conducted at the DLH. We screened all eligible blood bags recorded in 2022 at BTC for major TTIs using standard tests. The medical records of donors were extracted, and data of interest were reported using data collection sheets. All data of

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interest included age, gender, occupation, marital status, educational level, blood donation strategy, and donor type.

2.2. Study site

The study was conducted at the BTC of DLH located in the town of Douala, Littoral Region, Cameroon (Fig. 1). Douala is the economic capital and most populated town of Cameroon, with a high heterogeneity of ethnic groups which are mainly represented by *Duala*, *Bamileke*, and *Bassa* [24]. The BTC is structured into two sections: a sampling section where blood samples are collected (tubes and blood bags) and a screening section where tubes and blood bags are checked for conformity and screening of blood bags for TTIs. Blood samples collected in tubes are used to determine the blood group and haemoglobin levels. Both healthy and contaminated blood bags are labelled, and associated information is recorded in the respective databases. Contaminated blood bags are stored in a room dedicated to the incineration of non-qualified blood products, whereas healthy blood bags are distributed to different wards of the DLH. Medical doctors, nurses, clinical biologists, technicians, and non-medical constitute the staff of BTC.

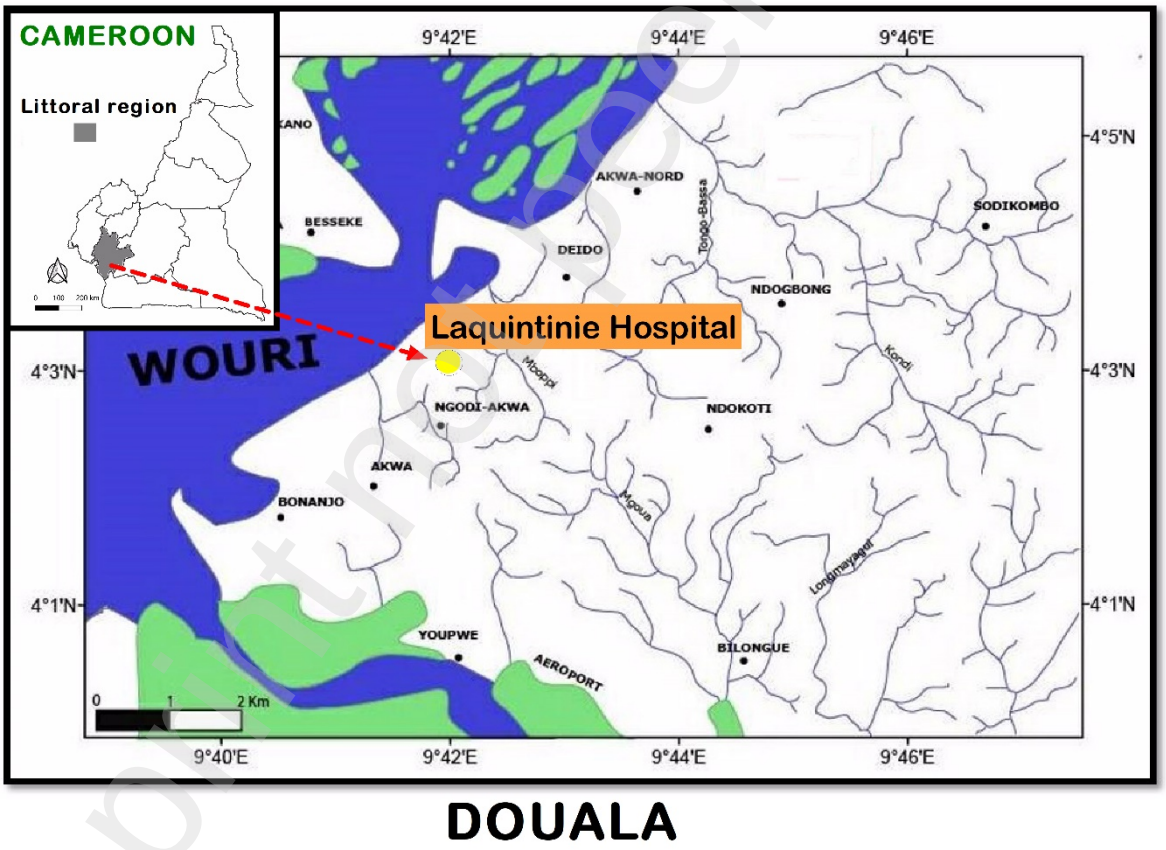


Fig. 1. Map showing the town of Douala, Littoral region.
The map was created using the QGIS software v3.10

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2.3. Population and sample size

In Cameroon, the main selection criteria for blood donors, regardless of culture, religion or socioeconomic status, including i) age between 18 and 65 years, ii) a body weight ≥ 50kg, iii) being recognized as suitable for donation by a doctor from the BTC, and iv) being in good health (e.g. no anaemia, normal blood pressure) (https://pnts-minsante.cm/accueil).

The sample size was computed using Lorentz's formula $N = [p \times (1 - p) \times Z^2]/d^2$, where N = sample size required, p = assumed prevalence of TTI, Z = statistic for the desired confidence level (1.96 for 95% confidence level), and d = accepted margin of error (0.05). Based on the latest study on TTIs in Douala [22], the maximum prevalence was 13.7%. Thus, the minimum sample size was estimated as N = 181.67 ≈ 182.

2.4. Screening for TTIs

Venous blood samples were collected in 5-mL vacutainers from each blood bag and stored at 4°C. Serum was separated by centrifugation at 3000 × g for 5 min and then used for further screening of TTIs within 24 hours of the blood collection. Screening and confirmation of viral infection at the BTC were based on the detection of viral antigens using rapid diagnostic tests (RDTs), ELISA, and hemagglutination assays as depicted in Fig. 2.

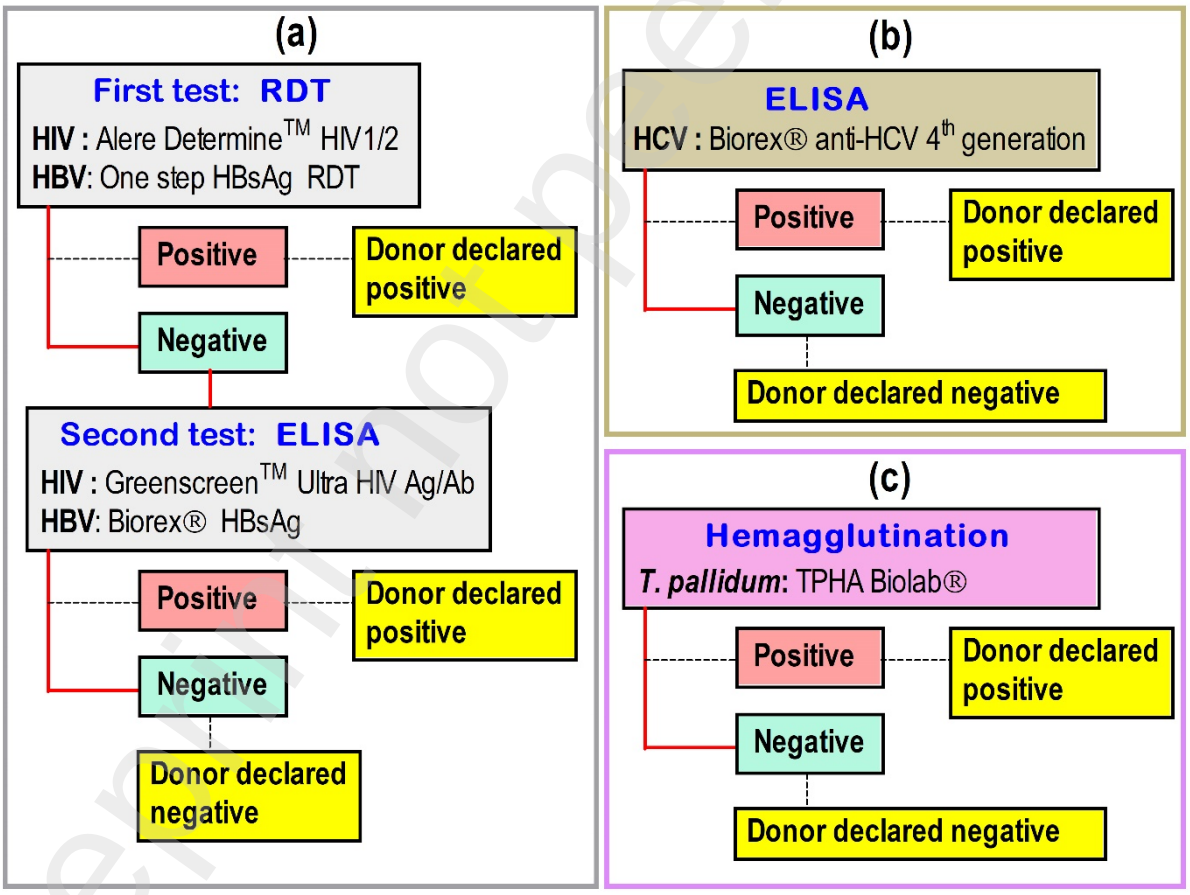


Fig. 2. Clinical algorithm used for screening of HBV and HIV (a), HCV (b) and *T. pallidum* (c).

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2.5. Quality assurance

Training was provided to the data collectors on the objectives, eligibility criteria, and data keying. A study sheet was generated to collect the data of interest. Principal investigators and senior members of the research group checked the collected data daily to maintain the accuracy and completeness of the data.

2.6. Operational definitions

Several definitions are used in this study [4,25].

- **TTI**: infection in which a pathogen – generally viruses, bacteria, and parasites – can be transmitted in donated blood through a transfusion to a recipient
- **Single infection**: presence of only one pathogen species in an individual.
- **Co-infection**: presence of at least two pathogen species in an individual.
- **Donor**: a person in defined good health condition who voluntarily donates blood or blood components.
- **First-time donor**: a donor whose blood or plasma is tested for the first time for infectious disease markers in a blood establishment.
- **Repeat donor**: Any individual who donates blood regularly without any break for a longer duration between donations.
- **Prevalence:** the rate of identified infection, including both past and present infections, at a specified point time in a defined population.

2.7. Ethical statements

The study was approved by the institutional review board of the University of Douala (No. 2950 CEI-UDo/01/2022/M). Administrative and research authorizations were also sought from the Delegation of Public Health for Littoral Region and Laquintinie Douala Hospital. No prospective phase was conducted during the study, thus, permission from all respective authorities was obtained to access raw data in the BTC. Patient-informed consent was not required in this case. Secondary data obtained from the electronic database was fully anonymized before researchers accessed the data.

2.8. Statistical analysis

Data collected during the study were keyed, coded, and verified for consistency in an Excel spreadsheet (Microsoft Office 2016, USA). Data were then performed using GraphPad v8.03 for Windows (GraphPad PRISM, IBM., California, USA), StatView v5.0 for Windows (SAS Institute, Inc., Chicago, Illinois, USA), and the statistical package for social sciences v16 for Windows (SPSS, IBM, Inc., Chicago, Illinois, USA) for statistical analyses. Quantitative variables were summarized as mean ± standard deviation (SD) while qualitative variables were summarized as frequency (*n*), percentage (%), and confidence interval at 95% (95%CI) in figures and tables where appropriate. Quantitative variables were tested for Gaussian distribution using the Agostino-Pearson test. Pearson's independence chi-square (χ^2) and Fisher's exact tests were used to compare proportions upon respect to Cochran's rule. Multiple correspondence analysis (MCA) was used to evaluate patterns of relationship between TTIs and patients' characteristics (i.e. gender, age, educational level, marital status, occupation, blood donation strategy, and type of blood donors) [26]. A Venn diagram was

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generated using the InteractiVenn tool to depict the different single infections and co-infections found in the study [27]. Univariate and multivariate logit models were used to identify determinants of TTIs and carriage of multiple pathogens. The association between independent variables and each TTI was quantified by crude and adjusted odds ratio (cOR and aOR), 95%CIs, and level of statistical significance. The independent variables included in the logit model were gender, age, educational level, marital status, occupation, blood donation, and type of blood donors. The goodness-of-fit of the final models was appraised by the Wald test and coefficient of multiple determination (R^2). A two-sided *p*-value < 0.05 was considered statistically significant for all analyses.

3. Results

3.1. Characteristics of blood donors

A total of 1,728 blood donors were included in the study; among them, 805 (46.6%) and 923 (53.4%) were first-time donors and repeat donors, respectively. The mean age ± SD of individuals was 31.0 ± 8.1 years (range 18 – 63 years) for first-time donors and 32.9 ± 8.9 years (range 15 – 62 years) for repeat donors, and the difference was statistically significant ($p < 0.0001$). As presented in Table 1, the sociodemographic profile of donors varied significant concerning the type of donor. Indeed, the proportion of males was lower in first-time donors as compared to that in repeat donors (84.1% vs. 90.6%, $p < 0.0001$). In contrast, the proportion of individuals who completed university studies was higher in first-time donors (40.5% *vs* 39.7%, $p < 0.0001$). In both groups, the bulk of donors were working in the informal sector in comparable proportions, but with a higher proportion of students in the first-time donor group (24.8% vs. 7.4%, $p = 0.0005$) (Table 1).

Table 1

Details of blood donors.

	First-time donors (*n* = 805)		**Repeat donors (*n* = 923)**		**Total (N = 1728)**			
Variables	*n*	%	*n*	%	*N*	%	**χ^2 (df)**	***p*-value**
Gender								
Females	128	15.9	87	9.4	215	12.4	105.3 (4)	**< 0.0001***
Males	677	84.1	836	90.6	1513	87.6		
Age (years)								
[15 - 25[	185	23.0	143	15.5	328	19.0	564.3 (12)	**< 0.0001***
[25 - 35[	381	47.3	436	47.2	817	47.3		
[35 - 45[	184	22.9	231	25.0	415	24.0		
≥ 45	55	6.8	113	12.2	168	9.7		
Educational level								
Primary	56	7.0	60	6.5	116	6.8	495.1 (8)	**< 0.0001***
Secondary	423	52.5	497	53.8	920	53.2		
University	326	40.5	366	39.7	692	40.0		

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Marital status								
Single	556	69.1	624	67.6	1180	68.3	182.6 (8)	**< 0.0001***
Married	245	30.4	294	31.9	539	31.2		
Divorced/Widow (er)	4	0.5	5	0.5	9	0.5		
Occupation								
Student	200	24.8	161	17.4	361	20.9	19.8 (4)	**0.0005***
Formal sector	163	20.2	214	23.2	377	21.8		
Informal sector	357	44.3	418	45.3	775	44.8		
Unemployed	57	7.1	73	7.9	130	7.5		
Other	28	3.5	57	6.2	85	4.9		
Blood donation								
Volunteer	121	15.0	153	16.6	274	15.9	128.5 (8)	**< 0.0001***
Family replacement	672	83.5	711	77.0	1383	80.0		
Remunerated	12	1.5	59	6.4	71	4.1		

Data are presented as frequency and percentage
Pearson's independence chi-square test was used to compare groups
χ^2: Decision variable of chi-square test, df: Degree of freedom
*Statistically significant at *p*-value < 0.05

3.2. Prevalence of TTIs among blood donors

The overall prevalence of TTIs was 19.6% (*n* = 336, 95%CI 17.8 – 21.5%) in this study. The most frequent TTI recorded in the study was HCV which was found at an overall prevalence of 9.3% (*n* = 161, 95%CI 8.0 – 10.8%) while HBV was less frequently found in blood samples at 3.4% (*n* = 58, 95%CI 2.6 – 4.3%) (Fig. 3a). TTIs were found to be either single infections (16.4%) or co-infections (3.2%), which were mainly represented by HCV + *T. pallidum* (*n* = 21, 1.2%) and HCV + HIV (*n* = 13, 0.8%) (Fig. 3b). Three types of co-infections were found in individuals and were represented by HBV + HCV + HIV (*n* = 2, 0.1%), HBV + HCV + *T. pallidum* (*n* = 2, 0.1%), and HBV + HIV + *T. pallidum* (*n* = 2, 0.05%) (Fig. 3b).

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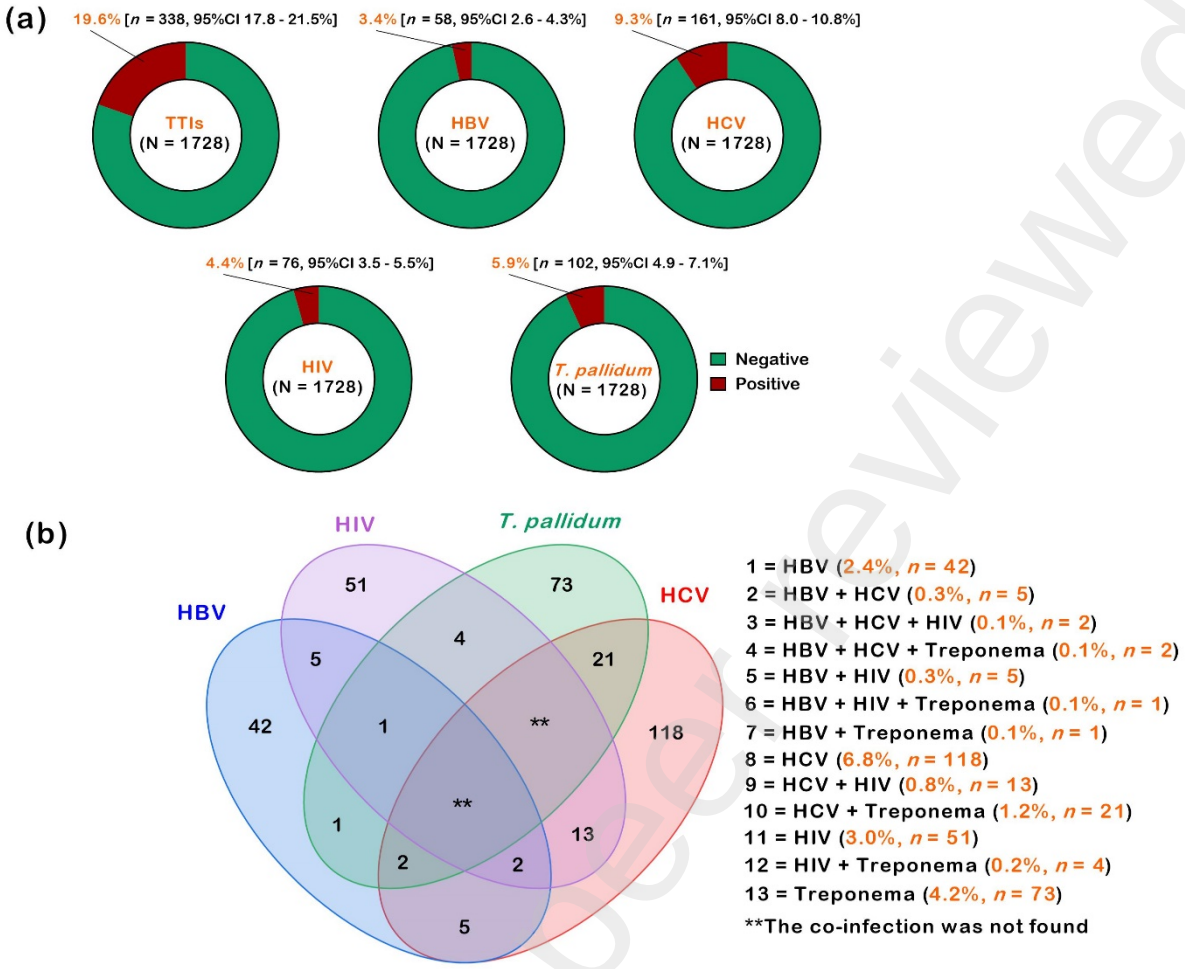


Fig. 3. Overall prevalence of each TTI (a), and Venn diagram of distribution of TTIs as either single infections or co-infections (b).
The area of each part of the Venn diagram is not proportional to the sample size
95%CI: Confidence interval at 95%, HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus, TTIs: Transfusion-transmissible infections

3.3. Prevalence of TTIs and associations of TTIs with donors' characteristics

As summarised in Table 2, no statistically significant association was found between HBV prevalence and blood donors' characteristics. The prevalence of HCV was higher in donors who had completed secondary studies (11.0%, $p = 0.03$) and in repeat blood donors (10.6%, $p = 0.04$). HIV infection was more frequently seen in remunerated blood donors (14.1%) compared to volunteers (6.9%) and family donors (3.4%), and the difference was statistically significant ($p < 0.0001$). Carriage of *T. pallidum* was significantly higher in those aged ≥ 45 years (14.3%, $p < 0.0001$) and in those having completed primary studies (8.6%, $p = 0.003$) (Table 2).

The distribution of different associations of TTIs was biased only by the donor's age and type of blood donation (Fig. 4 a - g). The proportion of mono-infection with either *T. pallidum* (11.3%), HIV (5.4%), or HBV (3.0%) was significantly higher in donors aged ≥ 45 years, while mono-infections with HCV (9.4%) were more frequently seen in donors aged 35 – 45 years

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(χ^2 = 84.32, df = 39, $p < 0.0001$) (Fig. 4b). The number of different TTIs in donors increased with age, ranging from 8 types of associations of TTIs in donors aged 15 – 35 years to 10 types of associations of TTIs in donors aged ≥ 45 years. Two co-infections, namely HBV + HCV + HIV (0.5%) and HBV + HCV + *T. pallidum* (0.5%) were only seen in donors aged 35 – 45 years. Likewise, HBV + HIV + *T. pallidum* (0.6%) and HBV + *T. pallidum* (0.6%), were only seen in donors aged ≥ 45 years (Fig. 4b). Regarding the type of blood donation, we noted that i) the proportion of mono-infection with HBV was higher in volunteer donors (4.4%), ii) the proportion of mono-infection with HCV was higher in family replacement donors (7.2%), and iii) the proportions of mono-infections with HIV and *T. pallidum* were higher in remunerated donors (9.9% and 5.6%, respectively) (Fig. 4f). A higher number of types of co-infections was found in family replacement donors (8 types of co-infections), while 5 and 4 co-infections were found in volunteer donors and remunerated donors, respectively. The co-infections HBV + HCV + HIV (0.1%), HBV + HCV + *T. pallidum* (0.1%), and HBV + HIV + *T. pallidum* (0.1%) were only recorded in family donors, while the co-infection HBV + *T. pallidum* (0.4%) was only seen in volunteer donors (Fig. 4f).

A high correlation was found between donors' characteristics and TTIs, as depicted in Fig. 5 a & 5b. Four clusters of variables were identified from MCA and included educational level, blood donation strategy, gender (cluster 1), occupation (cluster 2), marital status and age (cluster 3), and donor type, HBV, HIV, HCV, and *T. pallidum* (cluster 4) (Fig. 5a). For instance, first-time donors were more frequently aged 25 – 35 years and negative for HCV, HIV, and *T. pallidum*. In contrast, repeat donors were mostly aged 35 – 45 years or more than 45 years, and positive for HBV and HIV. In addition, an association was found between primary educational level, occupation (unemployed, informal sector), HCV, and *T. pallidum* (Fig. 5b).

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Table 2

Prevalence of TTIs by demographical details of the blood donors.

		TTIs		HBV		HCV		HIV		Treponema	
Variables	**N**	***n* (%)**	***p***	***n* (%)**	***p***	***n* (%)**	***p***	***n* (%)**	***p***	***n* (%)**	***p***
Gender											
Females	**215**	40 (18.6)	0.78	8 (3.7)	0.68	16 (7.4)	0.38	10 (4.7)	0.85	9 (4.2)	0.35
Males	**1513**	298 (19.7)		50 (3.3)		145 (9.6)		66 (4.4)		93 (6.1)	
Age (years)											
[15 - 25[	**328**	55 (16.8)	**0.001***	8 (2.4)	0.25	32 (9.8)	0.08	10 (3.0)	0.21	19 (5.8)	**< 0.0001***
[25 - 35[	**817**	143 (17.5)		24 (2.9)		65 (8.0)		36 (4.4)		37 (4.5)	
[35 - 45[	**415**	91 (21.9)		17 (4.1)		51 (12.3)		18 (4.3)		22 (5.3)	
≥ 45	**168**	49 (29.2)		9 (5.4)		13 (7.7)		12 (7.1)		24 (14.3)	
Educational level											
Primary	**116**	30 (25.9)	**0.003***	7 (6.0)	0.21	10 (8.6)	**0.03***	6 (5.2)	0.89	10 (8.6)	**0.003***
Secondary	**920**	198 (21.5)		27 (2.9)		101 (11.0)		39 (4.2)		67 (7.3)	
University	**692**	110 (15.9)		24 (3.5)		50 (7.2)		31 (4.5)		25 (3.6)	
Marital status											
Single	**1180**	229 (19.4)	0.57	34 (2.9)	0.21	118 (10.0)	0.26	55 (4.7)	0.41	67 (5.7)	0.11
Married	**539**	106 (19.7)		24 (4.5)		43 (8.0)		20 (3.7)		33 (6.1)	
Divorced/Widow (er)	**9**	3 (33.3)		0 (0.0)		0 (0.0)		1 (11.1)		2 (22.2)	
Occupation											
Student	**361**	59 (16.3)	0.19	5 (1.4)	0.11	32 (8.9)	0.18	12 (3.3)	0.84	21 (5.8)	0.48
Formal sector	**377**	66 (17.5)		17 (4.5)		26 (6.9)		17 (4.5)		17 (4.5)	
Informal sector	**775**	165 (21.3)		27 (3.5)		80 (10.3)		36 (4.6)		49 (6.3)	
Unemployed	**130**	28 (21.5)		4 (3.1)		17 (13.1)		7 (5.4)		7 (5.4)	
Other	**85**	20 (23.5)		5 (5.9)		6 (7.1)		4 (4.7)		8 (9.4)	
Blood donation											

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Volunteer	**274**	61 (22.3)	0.11	15 (5.5)	0.11	20 (7.3)	0.42	19 (6.9)	**< 0.0001***	17 (6.2)	0.88
Family replacement	**1383**	258 (18.7)		41 (3.0)		135 (9.8)		47 (3.4)		80 (5.8)	
Remunerated	**71**	19 (26.8)		2 (2.8)		6 (8.5)		10 (14.1)		5 (7.0)	
Type of blood donor											
First-time	**805**	142 (17.6)	0.06	28 (3.5)	0.79	63 (7.8)	**0.04***	29 (3.6)	0.16	40 (5.0)	0.12
Regular	**923**	196 (21.2)		30 (3.3)		98 (10.6)		47 (5.1)		62 (6.7)	

HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus, TTIs: Transfusion-transmissible infections

Data are presented as frequency and percentage

Fisher's exact test and Pearson's independence chi-square test were used to compare groups

χ^2: Decision variable of chi-square test, df: Degree of freedom

*Statistically significant at p-value < 0.05

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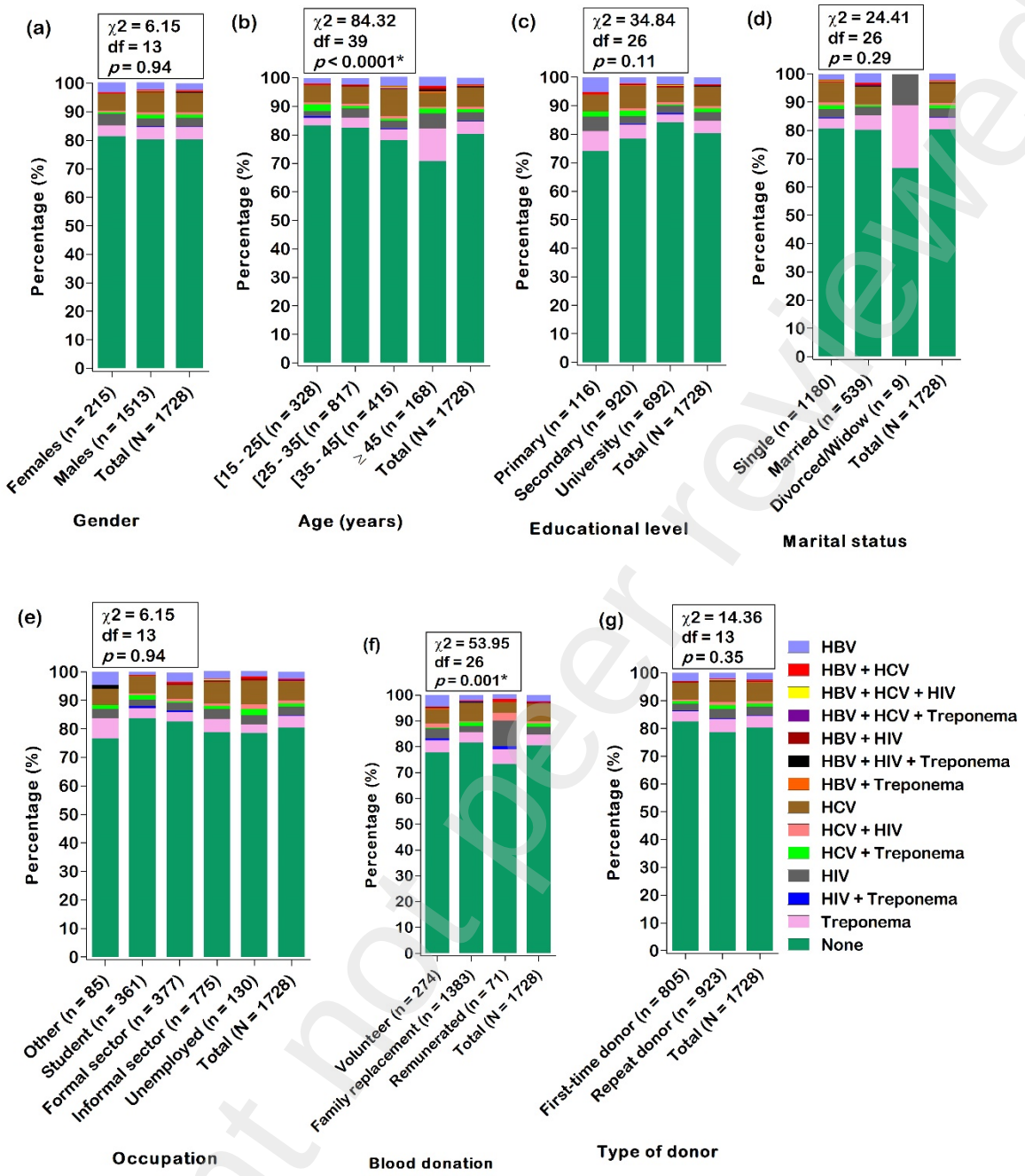


Fig. 4. Prevalence of associations of TTIs by gender (a), age (b), educational level (c), marital status (d), occupation (e), blood donation (f) and type of donor (g).

HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus, TTIs: Transfusion-transmissible infections

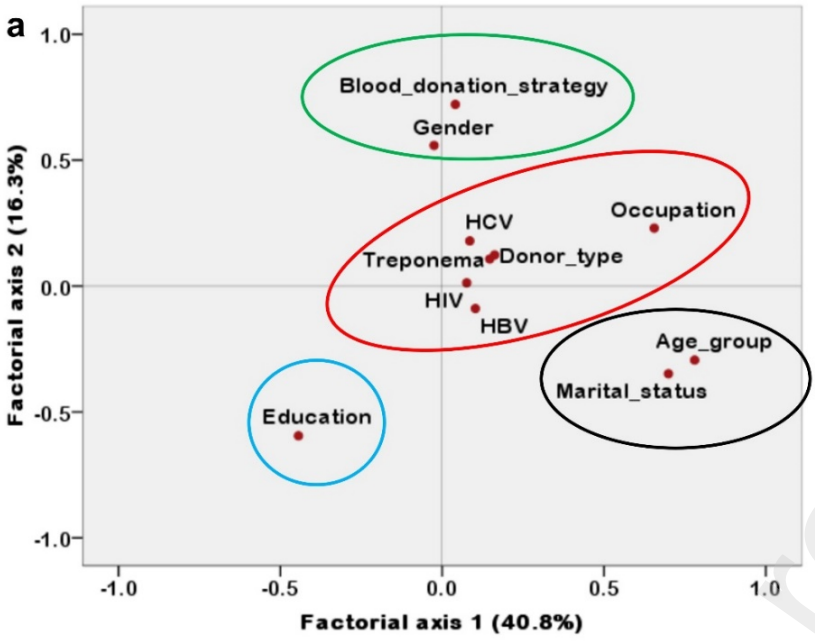
Pearson's independence chi-square test was used to compare groups

χ^2: Decision variable of chi-square test, df: Degree of freedom

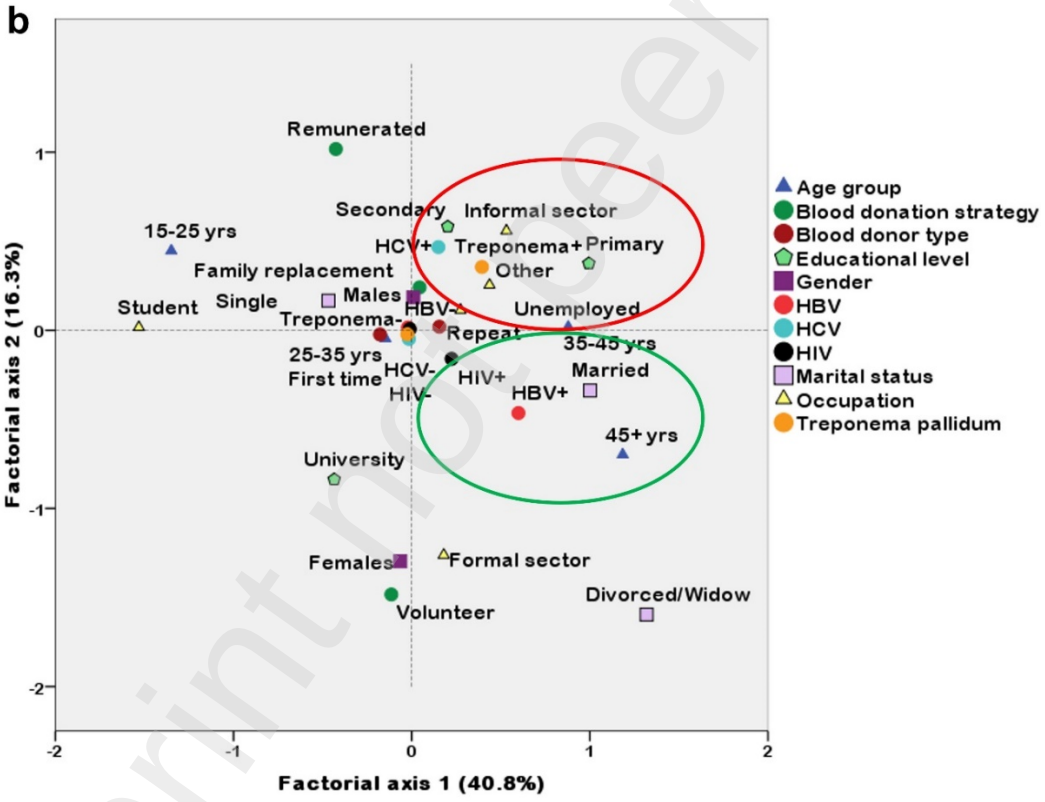
*Statistically significant at p-value < 0.05

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269 **Fig. 5**. Multiple correspondence analysis of the association between (a) variables, and (b)
270 TTIs and patients' characteristics.
271 HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus
272 In Figure 5a, four groups of association are identified between the variables analysed. The groups are
273 depicted by coloured circles
274 In Figure 5b, the modalities of the patients' characteristics are presented. For example, the green
275 circle depicts an association between the presence of HIV, HBV, being married, and aged 35 – 45
276 years or more than 45 years

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3.4. Determinants of TTIs among blood donors

On univariate logistic analysis, two determinants were found to be associated with the risk of TTIs among blood donors: age and educational level (Table 3). The risk of TTIs was two-fold (cOR = 2.04, 95%CI 1.31 – 3.18, *p* = 0.001) higher in donors aged ≥ 45 years compared to their younger counterparts aged 15 – 25 years. The risk of TTIs was reduced by 46% (cOR = 0.54, 95%CI 0.34 – 0.86, *p* = 0.009) in those with university educational level compared to those with primary educational level. Three determinants of TTIs, i.e., age, educational level, and type of blood donation, were identified in multivariate logistic analysis (Table 3). The odds of being infected with at least one pathogen were nearly doubled in donors aged 35 – 45 years (aOR = 1.62, 95%CI 1.01 – 2.59, *p* = 0.04) and nearly tripled in donors aged ≥ 45 years (aOR = 2.59, 95%CI 1.48 – 4.53, *p* = 0.0009) compared to those aged 15 – 25 years. The risk of TTIs was reduced by 44% (aOR = 0.56, 95%CI 0.34 – 0.94, *p* = 0.02) in those with university educational level compared to those with primary educational level. Finally, we noted that the risk of TTIs was reduced by 31% (aOR = 0.69, 95%CI 0.48 – 0.98, *p* = 0.03) in family replacement donors compared to their volunteer counterparts (Table 3).

Table 3

Univariate and multivariate logistic regression analysis of determinants of TTIs.

Variables	N	*n* (%)	cOR (95%CI)	*p*	aOR (95%CI)	*p*
Gender						
Females	**215**	40 (18.6)	1		1	
Males	**1513**	298 (19.7)	1.07 (0.74 - 1.55)	0.71	1.17 (0.78 - 1.76)	0.45
Age (years)						
[15 - 25[	**328**	55 (16.8)	1		1	
[25 - 35[	**817**	143 (17.5)	1.05 (0.75 - 1.48)	0.77	1.15 (0.78 - 1.71)	0.48
[35 - 45[	**415**	91 (21.9)	1.39 (0.96 - 2.02)	0.07	**1.62 (1.01 - 2.59)**	**0.04***
≥ 45	**168**	49 (29.2)	**2.04 (1.31 - 3.18)**	**0.001***	**2.59 (1.48 - 4.53)**	**0.0009***
Educational level						
Primary	**116**	30 (25.9)	1		1	
Secondary	**920**	198 (21.5)	0.79 (0.50 - 1.23)	0.30	0.87 (0.55 - 1.37)	0.55
University	**692**	110 (15.9)	**0.54 (0.34 - 0.86)**	**0.009***	**0.56 (0.34 - 0.94)**	**0.02***
Marital status						
Single	**1180**	229 (19.4)	1		1	
Married	**539**	106 (19.7)	1.02 (0.79 - 1.31)	0.89	0.76 (0.55 - 1.03)	0.08
Occupation						
Other	**361**	59 (16.3)	1		1	
Student	**377**	66 (17.5)	0.63 (0.36 - 1.13)	0.12	1.09 (0.57 - 2.09)	0.78
Formal sector	**775**	165 (21.3)	0.69 (0.39 - 1.22)	0.19	0.94 (0.51 - 1.72)	0.84
Informal sector	**130**	28 (21.5)	0.88 (0.52 - 1.49)	0.63	1.02 (0.59 - 1.76)	0.95
Unemployed	**85**	20 (23.5)	0.89 (0.46 - 1.71)	0.73	1.08 (0.55 - 2.14)	0.82
Blood donation						
Volunteer	**274**	61 (22.3)	1		1	
Family replacement	**1383**	258 (18.7)	0.80 (0.58 - 1.10)	0.17	**0.69 (0.48 - 0.98)**	**0.03***

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Remunerated	**71**	19 (26.8)	1.28 (0.70 - 2.32)	0.42	1.02 (0.54 - 1.94)	0.94
Type of blood donor						
First-time	**805**	142 (17.6)	1		1	
Repeat	**923**	196 (21.2)	1.26 (0.99 - 1.60)	0.06	1.15 (0.89 - 1.47)	0.28

95%CI: Confidence interval at 95%, cOR: Crude odds ratio, aOR: Adjusted odds ratio, TTIs: Transfusion-transmissible infections
Univariate and multivariate logistic regression was used to identify determinants of TTIs
*Statistically significant at $p < 0.05$

3.5. Determinants of TTIs with HBV, HCV, HIV or T. pallidum

Based on multivariate logistic analysis, only occupation was identified as a determinant of HBV infection, while marital status and donor type were determinants of HCV infection (Table 4). The risk of HBV infection was reduced by 78% (aOR = 0.22, 95%CI 0.05 – 0.89, p = 0.03) in students. Married donors were less at risk of HCV infection (aOR = 0.64, 95%CI 0.42 – 0.98, p = 0.04) compared to singles (Table 3). In contrast, the risk of HCV was increased by 1.41 times (95%CI 1.01 – 1.99, p = 0.04) in repeat donors compared to first-time donors. The age of donors and type of blood donation were associated with HIV infection (Table 4). Blood donors aged ≥ 45 years were more than three times more at risk of HIV compared to those aged 15 – 25 years (aOR = 3.17, 95%CI 1.08 – 9.28, p = 0.03). In contrast, the risk of HIV infection was reduced by 51% (aOR = 0.49, 95%CI 0.26 – 0.91, p = 0.02) in family replacement donors compared to volunteer donors. Regarding *T. pallidum* infection, two determinants were identified, and these included age and educational level. The risk of *T. pallidum* infection was ~5 times higher in donors aged ≥ 45 years compared to those aged 15 – 25 years (aOR = 4.94, 95%CI 2.05 – 11.92, p = 0.004). Again, this risk was reduced by 70% (aOR = 0.30, 95%CI 0.13 – 0.71, p = 0.006) in donors with university educational level compared to those with primary educational level (Table 4).

3.6. Determinants of single and multiple carriage of transfusion-transmitted pathogens

We stratified the risk analysis of TTIs based on the number of pathogens (i.e., one pathogen and more than one pathogen), as summarised in Table 5. The findings indicated that the risk of infection with only one pathogen was significantly modulated by age and educational level based on both univariate and multivariate logistic regression analyses. The risk of carrying only one pathogen was higher in donors aged 35 – 45 years (aOR = 1.78, 95%CI 1.06 – 2.99, p = 0.02) and nearly tripled in donors aged ≥ 45 years (aOR = 2.71, 95%CI 1.47 – 4.99, p = 0.001) compared to those aged 15 – 25 years. Donors with a university educational level had 47% fewer chances (aOR = 0.53, 95%CI 0.31 – 0.91, p = 0.02) of being infected with one pathogen compared to those with a primary educational level (Table 6). Besides, only donor type and marital status were associated with multiple carriage of pathogens in univariate and multivariate analyses, respectively. Indeed, the risk of multiple carriage of pathogens was nearly doubled in repeat blood donors (cOR = 1.82, 95%CI 1.03 – 3.24, p = 0.04) compared to first-time blood donors. In contrast, this risk was reduced by 62% (aOR = 0.38, 95%CI 0.17 – 0.84, p = 0.01) in married women as compared to singles (Table 6).

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335 **Table 4**

336 Univariate and multivariate logistic regression analysis of determinants of HBV and HCV infections.

	HBV				HCV			
Variables	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***
Gender								
Females	1		1		1		1	
Males	0.88 (0.41 - 1.89)	0.75	1.02 (0.45 - 2.33)	0.96	1.32 (0.77 - 2.26)	0.31	1.21 (0.68 - 2.16)	0.51
Age (years)								
[15 - 25[	1		1		1		1	
[25 - 35[	1.21 (0.54 - 2.72)	0.64	0.71 (0.29 - 1.78)	0.47	0.80 (0.51 - 1.25)	0.32	0.91 (0.55 - 1.52)	0.72
[35 - 45[	1.71 (0.73 - 4.01)	0.22	0.85 (0.30 - 2.39)	0.76	1.30 (0.81 - 2.07)	0.28	1.71 (0.93 - 3.13)	0.08
≥ 45	2.26 (0.86 - 5.98)	0.09	1.07 (0.33 - 3.53)	0.91	0.78 (0.40 - 1.52)	0.46	1.19 (0.53 - 2.67)	0.67
Educational level								
Primary	1		1		1		1	
Secondary	0.47 (0.20 - 1.11)	0.08	0.52 (0.22 - 1.23)	0.14	1.31 (0.66 - 2.58)	0.40	1.28 (0.64 - 2.55)	0.51
University	0.56 (0.24 - 1.33)	0.18	0.61 (0.23 - 1.61)	0.32	0.83 (0.41 - 1.68)	0.60	0.93 (0.43 - 2.00)	0.91
Marital status								
Single	1		1		1		1	
Married	1.57 (0.92 - 2.68)	0.09	1.11 (0.59 - 2.09)	0.74	0.78 (0.54 - 1.12)	0.18	**0.64 (0.42 - 0.98)**	**0.04***
Occupation								
Other	1		1		1		1	
Student	**0.22 (0.06 - 0.79)**	**0.02***	**0.22 (0.05 - 0.89)**	**0.03***	1.28 (0.52 - 3.17)	0.59	1.64 (0.61 - 4.35)	0.32
Formal sector	0.76 (0.27 - 2.11)	0.59	0.69 (0.23 - 2.05)	0.51	0.98 (0.39 - 2.45)	0.96	1.23 (0.47 - 3.17)	0.67
Informal sector	0.58 (0.22 - 1.54)	0.27	0.62 (0.23 - 1.69)	0.35	1.52 (0.64 - 3.59)	0.34	1.53 (0.64 - 3.66)	0.34
Unemployed	0.51 (0.13 - 1.95)	0.32	0.53 (0.13 - 2.12)	0.37	1.98 (0.75 - 5.25)	0.17	2.17 (0.80 - 5.87)	0.13
Blood donation								
Volunteer	1		1		1		1	

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Family replacement	**0.53 (0.29 - 0.97)**	**0.03***	0.56 (0.28 - 1.10)	0.09	1.37 (0.84 - 2.24)	0.20	1.11 (0.66 - 1.88)	0.69
Remunerated	0.50 (0.11 - 2.24)	0.37	0.59 (0.12 - 2.83)	0.51	1.17 (0.45 - 3.04)	0.74	0.82 (0.30 - 2.21)	0.69
Type of blood donor								
First-time	1		1		1		1	
Repeat	0.93 (0.55 - 1.57)	0.79	0.83 (0.48 - 1.43)	0.50	**1.40 (1.00 - 1.95)**	**0.04***	**1.41 (1.01 - 1.99)**	**0.04***

95%CI: Confidence interval at 95%, cOR: Crude odds ratio, aOR: Adjusted odds ratio, HBV: Hepatitis B virus, HCV: Hepatitis C virus
Univariate and multivariate logistic regression analysis was used to identify determinants of viral hepatitis infections among blood donors
*Statistically significant at $p < 0.05$

Table 5

Univariate and multivariate logistic regression analysis of determinants of HIV and syphilis infections.

	HIV				***Treponema pallidum***			
Variables	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***
Gender								
Females	1		1		1		1	
Males	0.94 (0.47 - 1.85)	0.85	1.18 (0.55 - 2.54)	0.67	1.50 (0.74 - 3.02)	0.26	1.82 (0.79 - 4.20)	0.16
Age (years)								
[15 - 25[	1		1		1		1	
[25 - 35[	1.47 (0.72 - 2.99)	0.29	1.45 (0.66 - 3.19)	0.36	0.77 (0.44 - 1.36)	0.37	1.12 (0.57 - 2.19)	0.74
[35 - 45[	1.44 (0.66 - 3.17)	0.36	1.80 (0.70 - 4.62)	0.22	0.91 (0.48 - 1.71)	0.77	1.39 (0.61 - 3.18)	0.44
≥ 45	**2.45 (1.03 - 5.79)**	**0.04***	**3.17 (1.08 - 9.28)**	**0.03***	**2.71 (1.44 - 5.11)**	**0.002***	**4.94 (2.05 - 11.92)**	**0.004***
Educational level								
Primary	1		1		1		1	
Secondary	0.81 (0.34 - 1.96)	0.64	1.11 (0.42 - 2.93)	0.84	0.83 (0.42 - 1.67)	0.61	0.90 (0.44 - 1.85)	0.77
University	0.86 (0.35 - 2.11)	0.74	1.24 (0.43 - 3.58)	0.69	**0.40 (0.19 - 0.85)**	**0.01***	**0.30 (0.13 - 0.71)**	**0.006***
Marital status								
Single	1		1		1		1	

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Married	0.79 (0.47 - 1.33)	0.37	0.58 (0.31 - 1.07)	0.08	1.08 (0.70 - 1.67)	0.72	0.76 (0.44 - 1.30)	0.32
Occupation								
Other	1		1		1		1	
Student	0.70 (0.22 - 2.21)	0.53	0.92 (0.26 - 3.28)	0.90	0.59 (0.25 - 1.39)	0.23	1.68 (0.60 - 4.68)	0.32
Formal sector	0.96 (0.31 - 2.92)	0.94	1.05 (0.33 - 3.40)	0.93	0.45 (0.19 - 1.09)	0.08	0.90 (0.34 - 2.38)	0.83
Informal sector	0.99 (0.34 - 2.84)	0.98	1.26 (0.43 - 3.71)	0.68	0.65 (0.30 - 1.42)	0.28	0.83 (0.35 - 1.94)	0.67
Unemployed	1.15 (0.33 - 4.06)	0.83	1.57 (0.43 - 5.75)	0.49	0.55 (0.19 - 1.57)	0.26	0.78 (0.25 - 2.42)	0.67
Blood donation								
Volunteer	1		1		1		1	
Family replacement	**0.47 (0.27 - 0.82)**	**0.007***	**0.49 (0.26 - 0.91)**	**0.02***	0.93 (0.54 - 1.59)	0.79	0.65 (0.35 - 1.21)	0.17
Remunerated	2.20 (0.97 - 4.97)	0.05	2.26 (0.90 - 5.65)	0.08	1.15 (0.41 - 3.22)	0.80	0.70 (0.23 - 2.10)	0.52
Type of blood donor								
First-time	1		1		1		1	
Repeat	1.44 (0.89 - 2.30)	0.13	1.11 (0.67 - 1.83)	0.69	1.38 (0.91 - 2.07)	0.13	1.24 (0.81 - 1.91)	0.32

95%CI: Confidence interval at 95%, cOR: Crude odds ratio, aOR: Adjusted odds ratio, HIV: Human immunodeficiency virus

Univariate and multivariate logistic regression analysis was used to identify determinants of HIV and *T. pallidum* infections among blood donors

*Statistically significant at $p < 0.05$

Table 6

Determinants of multiple carriage of transfusion-transmissible pathogens.

	One pathogen				More than one pathogen			
Variables	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***	**cOR (95%CI)**	***p***	**aOR (95%CI)**	***p***
Gender								
Females	1		1		1		1	
Males	0.96 (0.66 - 1.41)	0.84	1.02 (0.67 - 1.56)	0.93	2.45 (0.76 - 7.93)	0.14	2.84 (0.84 - 9.66)	0.09
Age (years)								

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[15 - 25[	1		1		1		1	
[25 - 35[	1.23 (0.84 - 1.79)	0.29	1.34 (0.87 - 2.08)	0.18	0.55 (0.27 - 1.11)	0.09	0.60 (0.27 - 1.35)	0.22
[35 - 45[	**1.60 (1.06 - 2.42)**	**0.02***	**1.78 (1.06 - 2.99)**	**0.02***	0.78 (0.36 - 1.69)	0.53	1.19 (0.45 - 3.12)	0.73
≥ 45	**2.29 (1.41 - 3.72)**	**0.008***	**2.71 (1.47 - 4.99)**	**0.001***	1.31 (0.54 - 3.21)	0.55	2.58 (0.82 - 8.10)	0.11
Educational level								
Primary	1		1		1		1	
Secondary	0.72 (0.45 - 1.15)	0.21	0.81 (0.50 - 1.30)	0.38	1.35 (0.41 - 4.49)	0.62	1.38 (0.41 - 4.69)	0.60
University	**0.51 (0.31 - 0.83)**	**0.006***	**0.53 (0.31 - 0.91)**	**0.02***	0.84 (0.24 - 2.92)	0.78	0.82 (0.21 - 3.12)	0.77
Marital status								
Single	1		1		1		1	
Married	1.12 (0.85 - 1.47)	0.41	0.84 (0.61 - 1.17)	0.31	0.56 (0.29 - 1.10)	0.09	**0.38 (0.17 - 0.84)**	**0.01***
Occupation								
Other	1		1		1		1	
Student	0.57 (0.31 - 1.05)	0.07	1.04 (0.53 - 2.06)	0.91	1.18 (0.26 - 5.47)	0.83	1.63 (0.32 - 8.41)	0.56
Formal sector	0.64 (0.35 - 1.16)	0.14	0.83 (0.44 - 1.57)	0.57	1.15 (0.25 - 5.31)	0.86	1.94 (0.40 - 9.40)	0.41
Informal sector	0.84 (0.48 - 1.46)	0.54	0.95 (0.54 - 1.69)	0.86	1.23 (0.28 - 5.32)	0.79	1.54 (0.35 - 6.76)	0.57
Unemployed	0.74 (0.37 - 1.50)	0.41	0.86 (0.41 - 1.78)	0.68	2.23 (0.45 - 11.07)	0.33	3.20 (0.63 - 16.30)	0.16
Blood donation								
Volunteer	1		1		1		1	
Family replacement	0.81 (0.58 - 1.14)	0.22	0.71 (0.49 - 1.04)	0.08	0.76 (0.37 - 1.54)	0.44	0.57 (0.26 - 1.25)	0.16
Remunerated	1.20 (0.63 - 2.31)	0.57	1.03 (0.51 - 2.06)	0.93	1.64 (0.49 - 5.43)	0.42	0.98 (0.27 - 3.53)	0.97
Type of blood donor								
First-time	1		1		1		1	
Repeat	1.18 (0.91 - 1.52)	0.21	1.08 (0.83 - 1.41)	0.57	**1.82 (1.03 - 3.24)**	**0.04***	1.61 (0.89 - 2.92)	0.12

95%CI: Confidence interval at 95%, cOR: Crude odds ratio, aOR: Adjusted odds ratio, HIV: Human immunodeficiency virus

Univariate and multivariate logistic regression analysis was used to identify determinants of multiple carriage of TTIs

*Statistically significant at $p < 0.05$

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4. Discussion

It is now well-proven that blood transfusion has been saving millions of lives around the globe, but this medical act is also an open window for the acquisition of several life-threatening pathogens, which are big challenges for blood donation systems in African countries. Here, we determined the prevalence and determinants of HBV, HCV, HIV, and *T. pallidum* among blood donors in Douala, Cameroon.

4.1. Characteristics of blood donors

The predominance of males in blood donation was noted in this study, which is consistent with previous studies in Cameroon and outside [21,28]. This finding is the result of traditional belief-related misconceptions in African settings, as males are thought to be healthier than women [29]. Other specifically female-related factors, such as cyclic blood losses during menstruation, pregnancy, and breastfeeding, also limit participation in blood donation [30].

Blood samples analysed were mainly from replacement donors, and this is consistent with previous investigators who reported high rates of replacement donors in Douala (89.5%) in the Littoral region [22] and Edéa (64.3%) in the east region [21]. Studies conducted in other Asian and African settings reported high rates of replacement donors [31]. In contrast, lower proportions of replacement donors were reported in Ethiopia [32]. Studies conducted in Tanzania, and Ethiopia reported that volunteer donors accounted for the bulk of donors [33,34]. Our finding points out that volunteer blood donation is still a big challenge to cover the national blood need, estimated at 400,000 blood bags (https://pnts-minsante.cm/accueil). Thus, it is crucial to promote strategies for ethical, non-remunerated blood donation in the country.

4.2. Prevalence of TTIs and co-infections

The overall prevalence of TTIs was 19.6%, which is close to estimates from previous studies in Edéa (21.2%) [22]. In contrast, our prevalence estimate is higher than that of studies conducted earlier in Douala (13.7%) [22], Yaoundé (13.4%) [35], and several settings including Ethiopia (4.68% and 12.4%) [33,36], Nigeria (14.9%) [37] and Tanzania (10.1%) [34], in which the same TTIs were targeted. The epidemiology of TTIs is shaped by a cocktail of factors related to the host (e.g., level of immunity, genetic polymorphisms, risk-increasing sexual behaviours), pathogens (e.g., circulating strains), and environment (e.g. control strategies), and spatial- and time-dependent variations could likely explain such discrepancies observed between the above-mentioned prevalence estimates. Differences in study design, sampling statistical fluctuations, and screening assays have also added heterogeneity to estimates.

Co-infections accounted for 3.2% of all infection cases in this study. This finding is not surprising as these pathogens share the same routes of transmission and some risk factors such as the number of sexual partners or resorting to blood transfusion [18,22]. Again, co-infections are epidemiologically and clinically important, as interactions between different pathogens could be favourable to one another (e.g., increased transmissibility and/or virulence); and thus, complicate the treatment of co-infected individuals because of possible drug-drug interaction-related severe adverse effects [38]. Several systematic reviews found that HCV/HIV co-infected individuals have a higher risk of faster progression to cirrhosis/histological fibrosis, and decompensated liver disease than individuals with HCV

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infection only [39,40], thereby suggesting that HIV infection could facilitate HCV infection directly or indirectly.

4.3. Determinants of TTIs

Several determinants of TTIs were identified in this study. A higher educational level was associated with a reduced risk of TTIs, and this finding was also reported elsewhere [33]. The positive impact of higher educational levels has also been demonstrated for other common major infectious diseases in Cameroon and the current COVID-19 pandemic [11,41]. People with higher education are able to better understand health messages from sources of information (i.e. television, radio, health campaigns), and thus, are more willing to change their behaviours, and adopt and/or implement TTIs control strategies. Consisting with previous studies [36], we also found that older donors were more at risk of TTIs. Several observations and assumptions could explain this result. The proportion of individuals with a high educational level was lower among donors aged 35 – 45 years compared to those aged 15 – 25 years. As above discussed, a high educational level is a key determinant of awareness and compliance with health prevention strategies.

Students were less at risk of HBV infections in this study, and this finding is supported by that from previous studies in Ethiopia [36]. The Government of Cameroon actively implemented and scaled up national campaigns and awareness strategies towards the prevention of sexually transmitted diseases in youths this last decade. The current awareness strategies exploit a large set of communication tools such as social media and phone applications which are mainly used by youths. Students have access more easily to health messages via these tools than their older peers. Married donors had less chance of being infected with HCV, and this could be due to a limited number of sexual partners as compared to singles.

In this study, being a replacement donor was associated with a reduced risk of HIV infection. Contradictory finding was found among Tanzanian blood donors [34]. We also found that older age was a risk factor for HIV infections, and a similar finding was reported earlier [34,36]. Also, this is consistent with national epidemiological data that show a higher HIV burden in older individuals, with prevalence ranging from ~0.2% in 0-14 years to ~3.4% in 15-49 years [42,43]. The risk of HIV infection was reduced in replacement donors, which is not in line with findings from an Ethiopian study [44]. It is possible that replacement donors, being family members/relatives of the recipient, are more aware of the infectious risks of blood transfusion, and thus, could adopt better health behaviours towards TTIs to preserve the health of the recipient.

Syphilis risk was strongly associated with poorer educational level and older age. As above discussed, persons with higher educational levels are more aware of health messages and the dangerousness of diseases, and thus, are more prompt to adopt good attitudes and behaviours towards the prevention and control of diseases. A similar association between low educational level and increased risk of syphilis was recently found among Chinese blood donors [45]. We found that older donors were more at risk of *T. pallidum* infection, which is consistent with what is known in the literature [45–47]. One study conducted in Ghana did not find any association between age and *T. pallidum* infection [48]. Previous studies reported other determinants of *T. pallidum* infection in blood donors - e.g. tattoo, multiple sexual partners, occupation, living area, donor type, and razor reuse [49,50] - which can outline geographical variations in determinants of this infection.

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5. Conclusion

We determined the prevalence and determinants of HBV, HCV, HIV, and *T. pallidum* pathogens among blood donors in Douala, Cameroon. The results indicated a high prevalence of these pathogens among donors, especially among those aged ≥ 45 years old and with a low educational level. TTIs were mainly represented as single infections, with a few donors carrying multiple infections. Older age was associated with a higher risk of TTIs, while a high educational level and being a family donor were protective factors against TTIs. Similar patterns of association were found between these three determinants and the risk of HIV and *T. pallidum*. The risk of carriage of multiple pathogen species was reduced in married individuals. It is crucial to improve the awareness of communities towards the promotion of blood bank activity to increase the number of repeated voluntary donors. Taken together, our findings will contribute to the ongoing efforts of the government of Cameroon to improve transfusion-related systems and strategies at both regional and national levels in Cameroon.

Abbreviations

95%CI: Confidence interval at 95%, aOR: Adjusted odds ratio, BTC: Blood transfusion centre, COVID-19: Coronavirus disease 2019, cOR: Crude odds ratio, df: Degree of freedom, DLH: Douala Laquintinie hospital, HBV: Hepatitis B virus, HCV: Hepatitis C virus, HIV: Human immunodeficiency virus, MCA: Multiple correspondence analysis, RDT: Rapid diagnostic test, SD: Standard deviation, SPSS: The statistical package for social sciences, TTI: Transfusion-transmittable infection, USA: The United States of America, WHO: World Health Organization

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data for this study may be made available on suitable request from the corresponding authors.

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References

[1] Carson JL, Stanworth SJ, Dennis JA, Trivella M, Roubinian N, Fergusson DA, et al. Transfusion thresholds for guiding red blood cell transfusion. Cochrane Database Syst Rev 2021;2021:CD002042. https://doi.org/10.1002/14651858.CD002042.pub5.

[2] Bloch EM, Vermeulen M, Murphy E. Blood transfusion safety in Africa. A literature review of infectious disease and organizational challenges. Transfus Med Rev 2013;26:164–80. https://doi.org/10.1016/j.tmrv.2011.07.006.Blood.

[3] Roberts DJ, Field S, Delaney M, Bates I. Problems and approaches for blood transfusion in the developing countries. Hematol Oncol Clin North Am 2016;30:477–95. https://doi.org/10.1016/j.hoc.2015.11.011.

[4] Bihl F, Castelli D, Marincola F, Dodd RY, Brander C. Transfusion-transmitted infections. J Transl Med 2007;5:25. https://doi.org/10.1186/1479-5876-5-25.

[5] Busch MP, Bloch EM, Kleinman S. Prevention of transfusion-transmitted infections. Blood 2019;133:1854–64. https://doi.org/10.1182/blood-2018-11-833996.

[6] Dean CL, Wade J, Roback JD. Transfusion-transmitted infections: An update on product screening, diagnostic techniques, and the path ahead. J Clin Microbiol 2018;56:e00352-18. https://doi.org/10.1128/JCM.00352-18.

[7] Inyimai SP, Ocan M, Wabwire B, Olupot-Olupot P. Asymptomatic Plasmodium parasites among adults in eastern Uganda: A case of donor blood screening at Mbale regional blood bank. J Trop Med 2018;2018:6359079. https://doi.org/10.1155/2018/6359079.

[8] World Health Organization. Blood safety: A strategy for the African region. Brazzaville: 2001.

[9] Services du Premier Ministre du Cameroun. Textes législatifs et réglementaires. Loi N° 2003/014 du 22 décembre 2003 portant sur la transfusion sanguine. 2003.

[10] Mbohou NC, Kojom Foko LP, Nyabeyeu Nyabeyeu H, Tonga C, Kouodjip Nono L, Kangam L, et al. Malaria screening at the workplace in Cameroon. PLoS One 2019;14:e0225219. https://doi.org/https://doi.org/10.1371/ journal.pone.0225219.

[11] Kojom Foko LP, Nolla NP, Nyabeyeu Nyabeyeu H, Lehman LG. Prevalence, patterns, and determinants of malaria and malnutrition in Douala , Cameroon : A cross-sectional community-based study. Biomed Res Int 2021;2021:5553344.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4869944

https://doi.org/https://doi.org/10.1155/2021/5553344.

[12] Bigna JJ, Amougou MA, Asangbeh SL, Kenne AM, Noumegni SRN, Ngo-malabo ET, et al. Seroprevalence of hepatitis B virus infection in Cameroon : a systematic review and meta-analysis. BMJ Open 2017;7:e015298. https://doi.org/10.1136/bmjopen-2016-015298.

[13] Bigna JJ, Amougou MA, Asangbeh SL, Kenne AM, Nansseu JR. Seroprevalence of hepatitis C virus infection in Cameroon: a systematic review and meta-analysis. BMJ Open 2017;7:e015748. https://doi.org/10.1136/bmjopen-2016-015748.

[14] Samje M, Fondoh VN, Nguefack-Tsague G, Kamalieuk LKJ, Mbanya D, Murphy EL, et al. Trends in serological markers of transfusion transmissible infections in blood donations at the Bamenda Hospital-based Blood Service, Cameroon. Transfus Clin Biol 2021;28:228–33. https://doi.org/10.1016/j.tracli.2021.05.011.

[15] Chetcha Chemegni B, Sack FN, Ndoumba A, Kenmalong Mboula LF, Sonkwa EL, Bertrand C, et al. Reasons for rejection of total blood pockets: Experience of the blood bank of Yaounde central hospital. J Blood Disord Transfus 1 J Blood Disord Transfus 2021;12:3.

[16] Mbanya D, Binam F, Kaptue L. Transfusion outcome in a resource-limited setting of Cameroon: A five-year evaluation. Int J Infect Dis 2001;5:70–3. https://doi.org/10.1016/S1201-9712(01)90028-1.

[17] Mbanya DN, Takam D, Ndumbet PM. Serological findings amongst first-time blood donors in Yaoundé, Cameroon: Is safe donation a reality or a myth? Transfus Med 2003;13:267–73. https://doi.org/10.1046/j.1365-3148.2003.00453.x.

[18] Koanga Mogtomo ML, Louandji Fomekong S, Fotso Kuate H, Ngono Ngane RA. Détection des agents infectieux dans les banques de sang de Douala (1995-2004). Cah Santé 2009;19:1–8.

[19] Noubouossie D, Tagny CT, Same-Ekobo A, Mbanya D. Asymptomatic carriage of malaria parasites in blood donors in Yaoundé. Transfus Med 2012;22:63–7. https://doi.org/10.1111/j.1365-3148.2011.01121.x.

[20] Koanga Mogtomo ML, Kojom Foko LP, Assokom Okoubalimba EV, Embolo Enyegue E, Ngono Ngane AR. High risk of transfusion-transmitted malaria (TTM) from student blood donors living in the town of Douala, Cameroon. J Clin Infect Dis Pract 2016;1:1. https://doi.org/10.4172/2476-213x.1000108.

[21] Noubiap JJN, Joko WYA, Nansseu JRN, Tene UG, Siaka C. Sero-epidemiology of human immunodeficiency virus, hepatitis B and C viruses, and syphilis infections among first-time blood donors in Edéa, Cameroon. Int J Infect Dis 2013;17:832–7. https://doi.org/10.1016/j.ijid.2012.12.007.

[22] Eboumbou Moukoko CE, Ngo Sack F, Essangui Same EG, Mbangue M, Lehman LG. HIV, HBV, HCV and *T. pallidum* infections among blood donors and transfusion-related complications among recipients at the Laquintinie hospital in Douala, Cameroon. BMC Hematol 2014;14:5. https://doi.org/10.1186/2052-1839-14-5.

[23] Fopa D, Candotti D, Tagny CT, Doux C, Mbanya D, Murphy EL, et al. Occult hepatitis B

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4869944

infection among blood donors from Yaoundé, Cameroon. Blood Transfus 2019;17:403–8. https://doi.org/10.2450/2019.0182-19.

[24] Loteri O, Moguem Sobgui AF, Kojom Foko LP, Ndeme Mboussi WS, Medi Sike C, Embolo Enyegue EL, et al. COVID-19 and comorbidities in Douala , Cameroon. Int J Trop Dis Heal 2022;43:21–38. https://doi.org/10.9734/IJTDH/2022/v43i1730658.

[25] WHO Expert Committee on Biological Standardization. Guidelines on estimation of residual risk of HIV, HBV or HCV infections via cellular blood components and plasma. Geneva: 2017. https://doi.org/WHO TRS N°1004.

[26] Gower J, Lubbe S, Roux N le. Multiple correspondence analysis. Underst Biplots 2010:365–403. https://doi.org/10.1002/9780470973196.ch8.

[27] Heberle H, Meirelles VG, da Silva FR, Telles GP, Minghim R. InteractiVenn: A web-based tool for the analysis of sets through Venn diagrams. BMC Bioinformatics 2015;16:169. https://doi.org/10.1186/s12859-015-0611-3.

[28] Wongjarupong N, Oli S, Sanou M, Djigma F, Koumare AK, Yonli AT, et al. Distribution and incidence of blood-borne infection among blood donors from regional transfusion centers in Burkina Faso: A comprehensive study. Am J Trop Med Hyg 2021;104:1577–81. https://doi.org/10.4269/ajtmh.20-0601.

[29] Nébié KY, Olinger CM, Kafando E, Dahourou H, Diallo S, Kientega Y, et al. Lack of knowledge of blood donors in Burkina Faso (West Africa); Potential obstacle to transfusion security. Transfus Clin Biol 2007;14:446–52. https://doi.org/10.1016/j.tracli.2007.12.005.

[30] Tagny C, Owusu-Ofori S, Mbanya D, Deneys V. The blood donor in sub-Saharan Africa: A review. Transfus Med 2010;20:1–10. https://doi.org/10.1111/j.1365-3148.2009.00958.x.

[31] Muhammad I, Saba N, Waheed U, Latif Z, Ullah W. Seroepidemiology and risk factors analysis of syphilis : A cross-sectional study from Peshawar, Pakistan. J Soc Obs Gynaecol Pak 2022;12:68–73.

[32] Degefa B, Gebreeyesus T, Gebremedhin Z, Melkamu G, Gebrekidan A, Hailekiros H, et al. Prevalence of hepatitis B virus, hepatitis C virus, and human immunodeficiency virus among blood donors of Mekelle blood bank, Northern Ethiopia: A three-year retrospective study. J Med Virol 2018;90:1724–9. https://doi.org/10.1002/jmv.25248.

[33] Heyredin I, Mengistie B, Weldegebreal F. Sero-prevalence of transfusion-transmittable infections and associated factors among blood donors in Eastern Ethiopia: an Institutional-based cross-sectional study. SAGE Open Med 2019;7:1–8. https://doi.org/10.1177/2050312119834468.

[34] Mremi A, Yahaya JJ, Nyindo M, Mollel E. Transfusion-transmitted infections and associated risk factors at the Northern zone blood transfusion center in Tanzania: A study of blood donors between 2017 and 2019. PLoS One 2021;16:e0249061. https://doi.org/10.1371/journal.pone.0249061.

[35] Tagny CT, Ndoumba A, Laperche S, Murphy E, Mbanya D. Reducing risks of transfusion-transmitted infections in a resource-limited hospital-based blood bank:

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4869944

601 the case of the Yaoundé University Teaching Hospital, Cameroon. ISBT Sci Res
602 2016;11:82–7. https://doi.org/10.1111/voxs.12287.

603 [36] Abdella S, Berheto TM, Tolera G, Belete W, Deressa T, Feleke A, et al. Sero-prevalence
604 of transfusion transmittable infections: HIV, Hepatitis B, C and *Treponema pallidum*
605 and associated factors among blood donors in Ethiopia: A retrospective study. PLoS
606 One 2020;15:e0241086. https://doi.org/10.1371/journal.pone.0241086.

607 [37] Okoroiwu HU, Okafor IM, Asemota EA, Okpokam DC. Seroprevalence of transfusion-
608 transmissible infections (HBV, HCV, syphilis and HIV) among prospective blood donors
609 in a tertiary health care facility in Calabar, Nigeria; An eleven years evaluation. BMC
610 Public Health 2018;18:645. https://doi.org/10.1186/s12889-018-5555-x.

611 [38] Macías J, Monge P, Mancebo M, Merchante N, Neukam K, Real LM, et al. High
612 frequency of potential interactions between direct-acting antivirals and concomitant
613 therapy in HIV/hepatitis C virus-coinfected patients in clinical practice. HIV Med
614 2017;18:445–51. https://doi.org/10.1111/hiv.12471.

615 [39] Graham C, Baden L, Yu E, Mrus J, Carnie J, Heeren T, et al. Influence of human
616 immunodeficiency virus infection on the course of hepatitis C virus infection: A meta-
617 analysis. Clin Infect Dis 2001;33:562–9. https://doi.org/10.3748/wjg.15.996.

618 [40] Deng LP, Gui XE, Zhang YX, Gao SC, Yang RR. Impact of human immunodeficiency virus
619 infection on the course of hepatitis C virus infection: A meta-analysis. World J
620 Gastroenterol 2009;15:996–1003. https://doi.org/10.3748/wjg.15.996.

621 [41] Moguem Soubgui AF, Embolo Enyegue EL, Kojom Foko LP, Ndeme Mboussi WS,
622 Deutou Hogoue G, Mbougang SP, et al. Epidemiological situation of SARS-CoV-2
623 infection in Douala, the most populated and highly heterogeneous town of Cameroon:
624 a post-vaccination update. Acta Trop 2023;241:106864.
625 https://doi.org/10.1016/j.actatropica.2023.106864.

626 [42] CAMPHIA. Cameroon population-based HIV impact assessment - CAMPHIA 2017.
627 Yaoundé: 2018.

628 [43] PEPFAR. Country Operational Plan (COP) 2021 - Strategic Direction Summary. 2021.

629 [44] Melku M, Ambachew S, Enawgaw B, Abebe M, Abebe Z, Deressa T, et al. Sero-
630 epidemiology and associated factors of HIV, HBV, HCV and syphilis among blood
631 donors in Ethiopia: a systematic review and meta-analysis. BMC Infect Dis
632 2021;21:778. https://doi.org/10.1186/s12879-021-06505-w.

633 [45] Zheng X, Ding W, Ling X, Shi J, Dong J, Yu G, et al. Prevalence of *Treponema pallidum*
634 antibody among volunteer blood donors in China. Can J Infect Dis Med Microbiol
635 2022;2022:1668703. https://doi.org/10.1155/2022/1668703.

636 [46] Kaur G, Kaur P. Syphilis testing in blood donors: An update. Blood Transfus
637 2015;13:197–204. https://doi.org/10.2450/2014.0146-14.

638 [47] Wu X, Guan Y, Ye J, Fu H, Zhang C, Lan L, et al. Association between syphilis
639 seroprevalence and age among blood donors in Southern China: An observational
640 study from 2014 to 2017. BMJ Open 2019;9:e024393.
641 https://doi.org/10.1136/bmjopen-2018-024393.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4869944

642 [48] Nkansah C, Serwaa D, Osei-Boakye F, Owusu-Ampomah R. Magnitude and trend of
643 HIV and Treponema pallidum infections among blood donors in Offinso-North District,
644 Ghana: a nine-year retrospective, cross-sectional study. Afr Health Sci 2022;22:465–
645 74. https://doi.org/10.4314/ahs.v22i1.55.

646 [49] Liu S, Luo L, Xi G, Wan L, Zhong L, Chen X, et al. Seroprevalence and risk factors on
647 syphilis among blood donors in Chengdu, China, from 2005 to 2017. BMC Infect Dis
648 2019;19:509. https://doi.org/10.1186/s12879-019-4128-7.

649 [50] Alharazi T, Alzubiery TK, Alcantara JC, Qanash H, Abdulrahman BS, Altayar MA, et al.
650 Prevalence of transfusion-transmitted infections (HCV, HIV, syphilis and malaria) in
651 blood Donors: A large-scale cross-sectional study. Pathogens 2022;11:726.

652

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